

SELECT with various clauses – GROUP BY, HAVING, ORDER BY

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database collegedb;
Query OK, 1 row affected (0.09 sec)

mysql> use collegedb;
Database changed
mysql> create table student(regno bigint primary key,stuname varchar(30),course varchar(20),marks int);
Query OK, 0 rows affected (0.15 sec)

mysql> insert into student values (191711341,'sonu','DBMS',85),(191711342,'pranathi','DBMS',90),(191711343,'roops','Java',75),(191711344,'abhighna','Java',88),(191711345,'tabassum','Python',80),(191711346,'krithika','Python',95);
Query OK, 6 rows affected (0.12 sec)
Records: 6 Duplicates: 0 Warnings: 0

mysql> select*from student;
+-----+-----+-----+-----+
| regno | stuname | course | marks |
+-----+-----+-----+-----+
| 191711341 | sonu | DBMS | 85 |
| 191711342 | pranathi | DBMS | 90 |
| 191711343 | roofs | Java | 75 |
| 191711344 | abhighna | Java | 88 |
| 191711345 | tabassum | Python | 80 |
| 191711346 | krithika | Python | 95 |
+-----+-----+-----+-----+
6 rows in set (0.06 sec)

mysql> select course,avg(marks) as averagemarks from student group by course;
+-----+-----+
| course | averagemarks |
+-----+-----+
| DBMS | 87.5000 |
| Java | 81.5000 |
| Python | 87.5000 |
+-----+-----+
3 rows in set (0.01 sec)

mysql> select course,avg(marks) as averagemarks from student group by course having avg(marks)>85;
+-----+-----+
| course | averagemarks |
+-----+-----+
| DBMS | 87.5000 |
+-----+-----+
```

```
mysql> select course,avg(marks) as averagemarks from student group by course having avg(marks)>85;
+-----+-----+
| course | averagemarks |
+-----+-----+
| DBMS | 87.5000 |
| Python | 87.5000 |
+-----+-----+
2 rows in set (0.01 sec)

mysql> select*from student order by marks asc;
+-----+-----+-----+-----+
| regno | stuname | course | marks |
+-----+-----+-----+-----+
| 191711343 | roofs | Java | 75 |
| 191711345 | tabassum | Python | 80 |
| 191711341 | sonu | DBMS | 85 |
| 191711344 | abhighna | Java | 88 |
| 191711342 | pranathi | DBMS | 90 |
| 191711346 | krithika | Python | 95 |
+-----+-----+-----+-----+
6 rows in set (0.05 sec)

mysql> select*from student order by marks desc;
+-----+-----+-----+-----+
| regno | stuname | course | marks |
+-----+-----+-----+-----+
| 191711346 | krithika | Python | 95 |
| 191711342 | pranathi | DBMS | 90 |
| 191711344 | abhighna | Java | 88 |
| 191711341 | sonu | DBMS | 85 |
| 191711345 | tabassum | Python | 80 |
| 191711343 | roofs | Java | 75 |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> select course, avg(marks) from student group by course;
+-----+-----+
| course | avg(marks) |
+-----+-----+
| DBMS | 87.5000 |
| Java | 81.5000 |
| Python | 87.5000 |
+-----+-----+
3 rows in set (0.00 sec)

mysql> select course, avg(marks) from student group by course having avg(marks)>85;
```

```
mysql> select course, avg(marks) from student group by course having avg(marks)>85;
```

course	avg(marks)
DBMS	87.5000
Python	87.5000

2 rows in set (0.00 sec)

```
mysql> select*from student order by marks asc;
```

regno	stuname	course	marks
191711343	roops	Java	75
191711345	tabassum	Python	80
191711341	sonu	DBMS	85
191711344	abhighna	Java	88
191711342	pranathi	DBMS	90
191711346	krithika	Python	95

6 rows in set (0.00 sec)

```
mysql> select*from student order by marks desc;
```

regno	stuname	course	marks
191711346	krithika	Python	95
191711342	pranathi	DBMS	90
191711344	abhighna	Java	88
191711341	sonu	DBMS	85
191711345	tabassum	Python	80
191711343	roops	Java	75

6 rows in set (0.00 sec)